Critical odd-degree tails behind exceptional ends

Human Compression project: L044 continuation

Throughout $G \le A_n$, $n \ge 12$, is transitive with connected string C-marking, all maximal omissions intransitive, and the alternating rank bound is available at smaller degrees at least twelve. This is a bounded reduction, not an exclusion of its final equality.

Lemma 1. *Suppose the first three generators are an actual supported six-point A_5 marking from source Table 2(2) or (3), allowing reversal of that displayed triple. If $r > \lfloor (n-1)/2 \rfloor$, then*

$$n = 2r \ge 18, \qquad H = G_{>2} = A_Z, \qquad |Z| = n - 5 \ge 13, \qquad \operatorname{rank}(H) = (|Z| - 1)/2.$$

Here Z is the complement of the first five vertices of the literal head, and $G_{>3}$ fixes its sixth vertex. All maximal omissions of H are intransitive on Z.

Proof. First settle the orientation literally. In the reversed row 2, its first two generators are (1 2)(5 6) and (2 3)(4 5); their orbits are the two triples $\{1,2,3\}$ and $\{4,5,6\}$. They have no fixed point in the head. Any commuting next generator therefore preserves the head, and every further generator centralizes the full head. There is no exit, contrary to global transitivity. Reversal of row 3 is conjugate to the original row by (1 6)(2 5)(3 4). Thus it suffices to use either displayed orientation, with no local reversal silently assumed harmless.

Use the exact accepted source-table labels: the first two generators are (1 2)(3 4) and (2 3)(4 5), and the third is either (1 2)(5 6) or (3 4)(5 6). The first two generate the natural D_{10} on $\{1,\dots,5\}$ and fix every other point. Its centralizer on those five points is trivial (centralizing the five-cycle gives a power of it, and its reflection inverts that power). Hence ρ_3, which commutes with the first two, fixes those five points. Every ρ_k, $k \ge 4$, centralizes the entire head A_5, so preserves its unique nontrivial orbit of size six and fixes it pointwise: the accepted six-point action is 2-transitive and has trivial centralizer. Thus H is supported on $Z = \Omega \setminus \{1,\dots,5\}$.

Any H-orbit in Z not containing vertex6 would be fixed pointwise by the head and preserved by all of G, contradicting transitivity. Therefore H is transitive on all of Z, and has faithful even C-marking of rank $d = r - 3$. Its endpoint omission fixes vertex6. A transitive maximal omission of H on Z would make the corresponding global omission already connect every point moved by its omitted generator; adjoining that generator could not join omission orbits. This contradicts transitivity of G.

Put $q = |Z| = n - 5$. Rank failure gives $d \ge (q-1)/2$. For $q < 12$ the accepted small supported-tail lemma leaves only five- or six-point groups, whereas $q \ge 7$ here. For $q \ge 12$ the accepted proper-primitive and anchored-imprimitive bounds exclude every proper transitive subgroup of A_Z at this rank. Thus $H = A_Z$. Smaller-degree alternating induction gives $d \le \lfloor (q-1)/2 \rfloor$. Comparing the two inequalities forces q odd and $n = 2r$. Since $q \ge 12$, this gives $q \ge 13$ and $n \ge 18$ as claimed. □

Lemma 2. *Suppose G_0 has the two orbits $\{a\}$ and B. In the same contradiction range,*

$$n = 2r \ge 14, \qquad G_0 = A_B, \qquad |B| = n - 1 \ge 13, \qquad \operatorname{rank}(G_0) = (|B| - 1)/2.$$

Moreover $G_{>1}$ is intransitive on its support.

Proof. The actual supported tail $G_0 = G_{>0}$ has degree $q = n-1$ and rank $d = r-1 \geq (q-1)/2$. Its evenness, IP and intransitive omissions follow as above. Its external extension supplies the endpoint anchor used by the complete small and large supported-tail bounds. Since $q \geq 11$, the small survivors of degrees5 and6 are impossible. The proper large cases are likewise excluded, so smaller-degree alternating induction gives the asserted full odd-degree alternating equality. The unique 0-crossing ab follows from the singleton orbit. Suppose $K = G_{>1}$ is transitive on its support X, of size x. It fixes a, b. If $X = \Omega \setminus \{a, b\}$, the complete accepted singleton-endpoint theorem already gives the global rank bound. Otherwise $x \leq n-3$ and $\mathrm{rank}(K) = r-2 = (n-4)/2 \geq (x-1)/2$. The actual supported-tail bounds apply again. Their small survivors have ranks two or three, impossible here since $n \geq 14$ gives $r-2 \geq 5$. The proper large cases are excluded, so $K = A_X$ and smaller-degree induction forces $x = n-3$. Its centralizer on X is trivial. Since ρ_0 commutes with K, it fixes X pointwise. On the three remaining points it contains $(a\ b)$ and hence is odd, a contradiction. Thus K is intransitive on its support. □

Limit. Neither lemma rules out its final odd-degree alternating marking. These are precise remaining configurations for the source Propositions 5.29–5.32 continuation in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical pages38–44. The literal head data are the accepted Table 2 transcript from physical page27. The source's end-subgroup strategy is retained; the actual five-fixed-point tail and its exact critical range are the present reduction using the complete paid small, anchored, primitive and induction interfaces. Both literal head orientations have been checked explicitly: reversed row 2 cannot extend and row 3 has the displayed reversing conjugation. All full small certificates and predecessor proofs remain paid. No new computation or formalization is used. This author proof has not yet received distinct review; no whole L044, original, question or priority credit is claimed.